Kai-Yuan Lin

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RESEARCH FOCUS

Visual analytics for understanding learned 3D representations and robot policies. Developing interactive tools that connect model internals with spatial evidence, actions, and task outcomes to support diagnosis and comparison.

EDUCATION

National Tsing Hua University 2026 - Present

M.S. student, Institute of Information Systems and Applications
Advisor: Prof. Shih-Hsuan Hung | Visualization and Modeling Lab

National Tsing Hua University 2026

B.S. in Computer Science

PUBLICATION

Kai-Yuan Lin, Aryabima Mandala Putra, Jui-Chi Lee, and Shih-Hsuan Hung.

Vis4GS: A Visual Analytic Tool for 3D Gaussian Splatting Reconstruction.

IEEE VIS 2026, Short Papers. Accepted. arXiv:2606.26985

RESEARCH EXPERIENCE

Visualization and Modeling Lab, NTHU 2024 - Present

Graduate and undergraduate research | Advisor: Prof. Shih-Hsuan Hung

Vis4VLA / VisPlatform 2026 - Present

Visual analytics for vision-language-action models | Ongoing research

- Developing a web-based workbench for GR00T N1.7 policies, including model architecture navigation, attention and activation capture, saved-run inspection, and episode replay.
- Built infrastructure for comparing compatible action heads on identical inputs, inspecting checkpoints, and recording simulator sessions and evaluation outcomes.
- Investigating how spatially grounded attention, action uncertainty, and deviations from successful trajectories can help localize robot-policy failures and guide follow-up experiments.

Vis4GS 2024 - 2026

Visual analytics for 3D Gaussian Splatting | C++, OpenGL, ImGui

- Co-developed an interactive system for diagnosing 3D reconstruction artifacts by linking rendered scenes to Gaussian properties, training history, and view coverage.
- Implemented Gaussian genealogy exploration to trace clone, split, and prune operations across training iterations, alongside interactive scene inspection and metric timelines.
- The four-view system integrates artifact scoring and multiscale densification trees; a comparative user study reported better usability and artifact understanding than the original 3DGS viewer.

HONORS AND AWARDS

Best Paper Award 2026
Computer Graphics Workshop (CGW) | Vis4GS

First Place, TSMC IT CareerHack Feb 2026
Group D: AI-Powered Legacy Code Refactoring

Bronze Award 2026
NTHU College of EECS Practical Project Competition | Vis4GS

Outstanding Award (優等) 2026
NTHU Computer Science Senior Project Competition | Vis4GS

CONFERENCE PRESENTATION

IEEE VIS 2026, Boston, USA Nov 13, 2026
Vis4GS: A Visual Analytic Tool for 3D Gaussian Splatting Reconstruction.
Short-paper talk in Big Data, Bigger Physics. Scheduled.

TEACHING EXPERIENCE

Teaching Assistant, Introduction to Programming (C++) Fall 2026
National Tsing Hua University | TAICA course

- Support programming lab demonstrations and student questions; help manage lab submission and makeup arrangements.

SELECTED ENGINEERING PROJECTS

AI-Assisted Scheduling Application Feb - May 2024
Lead Developer | Software Studio course | Dart, Flutter

- Developed an AI-assisted scheduling app and coordinated sprint planning and the team backlog.

FPGA Retro Game Nov 2023 - Jan 2024
Developer | Hardware Design and Lab course | Verilog

- Implemented and tested real-time graphics and game logic on an FPGA using Verilog.

TECHNICAL SKILLS

Programming: C++, Python, C, Dart, Verilog

Visualization and development: OpenGL, ImGui, Flutter, Git, Docker, Linux

Model analysis: Inference instrumentation, attention inspection, checkpoint comparison, robot-policy evaluation

LEADERSHIP

Events Head, The Holders Freshmen Orientation Team Mar - Aug 2023
Organized freshman orientation activities for more than 30 new students at NTHU.

LANGUAGES

English (TOEIC 990) | Mandarin (native) | Japanese (JLPT N2) | Tagalog (conversational)